# Titanic Dataset
## Exploratory Data Analysis

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| **Submitted by** | Gopika Ajith |
| **Internship** | Elevate Labs – Data Analytics Internship |
| **Task** | Day 5 – Exploratory Data Analysis (EDA) |
| **Dataset** | Titanic Passenger Dataset |
| **Tools Used** | Python, Pandas, NumPy, Matplotlib, Seaborn |
| **Platform** | Google Colab |

**GitHub Repository:** https://github.com/Gopika-Ajith/Day-5-EDA-Titanic

## 1. Dataset Overview

### Objective

Perform Exploratory Data Analysis (EDA) on the Titanic passenger dataset to extract meaningful insights using visual and statistical exploration techniques.

### Dataset Information

| Attribute | Detail |
|---|---|
| Source | datasciencedojo / titanic.csv (GitHub) |
| Records | 891 passenger entries |
| Features | 12 columns (numerical + categorical) |
| Target | Survived (0 = No, 1 = Yes) |

## 2. Missing Value Analysis

| Column | Missing Values | Handling Strategy |
|---|---|---|
| Age | 177 (19.9%) | Moderate – impute with median |
| Cabin | 687 (77.1%) | High – drop or encode presence |
| Embarked | 2 (0.2%) | Low – fill with mode |

## 3. Statistical Summary

Key descriptive statistics from df.describe():

| Metric | Age | Fare | Pclass | SibSp | Parch |
|---|---|---|---|---|---|
| Mean | 29.70 | 32.20 | 2.31 | 0.52 | 0.38 |
| Std Dev | 14.53 | 49.69 | 0.84 | 1.10 | 0.81 |
| Min | 0.42 | 0.00 | 1.00 | 0.00 | 0.00 |
| Max | 80.00 | 512.33 | 3.00 | 8.00 | 6.00 |
| Median (50%) | 28.00 | 14.45 | 3.00 | 0.00 | 0.00 |

## 4. Visualizations & Observations

### Visualization 1: Survival Distribution

```
sns.countplot(x='Survived', data=df)
```

- More passengers died (549) than survived (342).
- Survival distribution is imbalanced — ~38% survived.
- Class imbalance must be considered in modeling.

### Visualization 2: Survival by Gender

```
sns.countplot(x='Sex', hue='Survived', data=df)
```

- Female passengers had significantly higher survival rates (~74%).
- Most male passengers did not survive (~19% survival).
- Gender is the strongest individual predictor of survival.

### Visualization 3: Passenger Class Distribution

```
sns.countplot(x='Pclass', data=df)
```

- Third Class held the most passengers (491 out of 891).
- First Class: 216 passengers, Second Class: 184 passengers.
- Class distribution reflects socio-economic diversity onboard.

## Visualization 4: Age Distribution

```
sns.histplot(df['Age'].dropna(), bins=30)
```

- Most passengers were between 20 and 40 years old.
- Peak concentration around mid-20s.
- Very few passengers were elderly (above 60).

## Visualization 5: Fare Distribution

```
sns.histplot(df['Fare'], bins=30)
```

- Fare distribution is highly right-skewed.
- Majority of passengers paid low fares (below 50).
- Several high-value outliers exist (max = 512.33) indicating luxury cabins.

## Visualization 6: Age vs Passenger Class (Boxplot)

```
sns.boxplot(x='Pclass', y='Age', data=df)
```

- First Class passengers were generally older (median ~37 years).
- Third Class passengers were younger on average (median ~25 years).
- Outliers present in all classes.

## Visualization 7: Age vs Fare (Scatter Plot)

```
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
```

- No strong linear relationship between Age and Fare.
- Higher fare passengers had noticeably higher survival rates.
- Several fare outliers visible regardless of age.

## Visualization 8: Correlation Heatmap

```
sns.heatmap(df.select_dtypes(include=['number']).corr(), annot=True,
cmap='coolwarm')
```

- Pclass and Fare: strong negative correlation (–0.55).
- Survived and Fare: positive correlation (0.26).
- Survived and Pclass: negative correlation (–0.34).
- SibSp and Parch: moderate positive correlation (0.41).

## Visualization 9: Pairplot

```
sns.pairplot(df[['Age','Fare','Pclass','Survived']].dropna())
```

- Visual comparison of Age, Fare, Pclass, and Survival.
- Fare distribution shows clear outliers across all pairings.
- No strong linear relationships among most variable pairs.

## 5. Key Findings

- 891 passenger records across 12 features were analysed.
- Missing values detected: Age (177), Cabin (687), Embarked (2).
- Overall survival rate was approximately 38% — majority did not survive.
- Gender was the most influential survival factor: ~74% of females survived vs ~19% of males.
- Third Class carried the highest number of passengers but had the worst survival odds.
- Most passengers were aged 20–40; ticket fares were heavily right-skewed.
- First Class passengers were older on average than Third Class passengers.
- Strong negative correlation (–0.55) between Passenger Class and Fare.
- Higher-fare passengers survived more often, reinforcing the class-survival link.
- Correlation analysis confirmed Pclass and Fare as the strongest numerically related pair.

## 6. Conclusion

The Exploratory Data Analysis of the Titanic dataset successfully uncovered several important patterns related to passenger survival. Gender, passenger class, and ticket fare emerged as the most influential factors. Female passengers and those in higher classes had significantly better survival outcomes. The analysis identified missing values requiring treatment, highlighted outliers in the Fare variable, and confirmed meaningful correlations among numerical features. These findings provide a strong foundation for subsequent predictive modelling tasks.

## 7. Deliverables

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|---|---|
| **Titanic_EDA.ipynb** | Jupyter Notebook with full analysis |
| **titanic.csv** | Raw dataset used for analysis |
| **README.md** | Professional project documentation |
| **GitHub Repository** | https://github.com/Gopika-Ajith/Day-5-EDA-Titanic |
| **9 Visualizations** | With observations for each chart |
| **PDF Report** | This document |